AMAZON BUSINESS INTELLIGENCE & DEMAND FORECASTING PLATFORM

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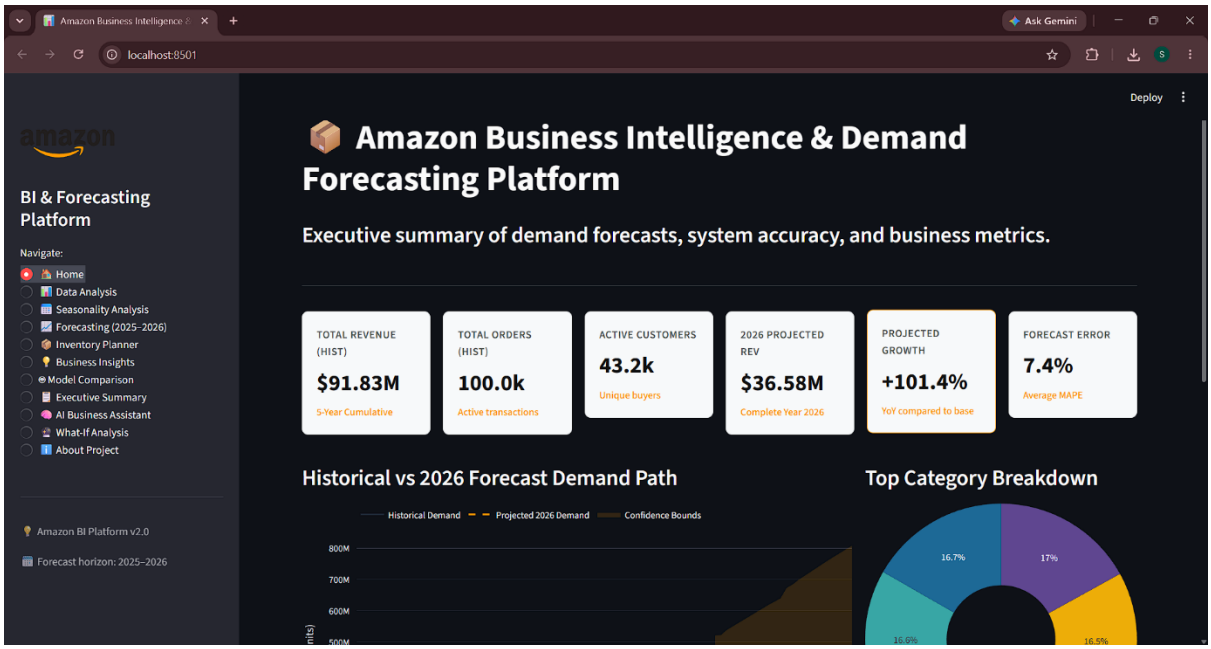
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1. Introduction

The Amazon Business Intelligence & Demand Forecasting Platform is an end-to-end analytics and forecasting application designed to transform historical Amazon transaction data into actionable business intelligence. The application combines Business Intelligence, Machine Learning, Time Series Forecasting, and Inventory Planning to support strategic business decisions.

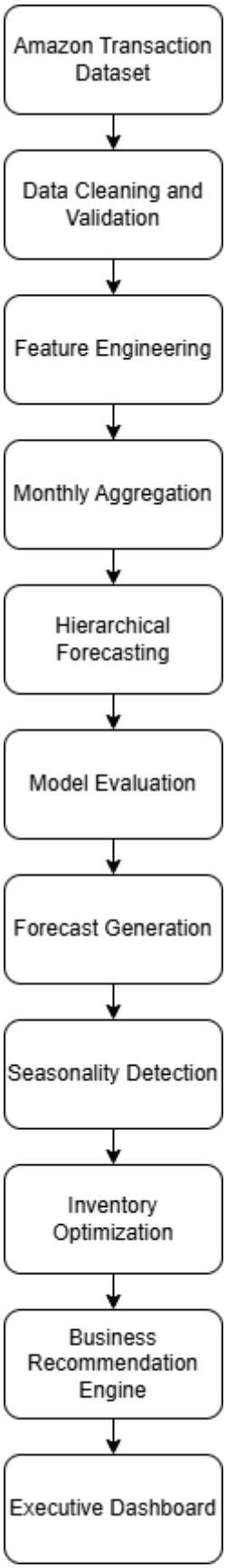
The primary objective of the application is to analyze five years of historical Transaction data (2020–2024), identify seasonal demand patterns, forecast future demand for 2025 and 2026, and provide inventory planning recommendations.

Unlike traditional reporting systems, the application not only visualizes historical data but also predicts future demand and converts predictions into business recommendations.



2. Application Workflow

The application follows a modular pipeline where each module performs a specific analytical task.



Each module is independent, allowing the system to be easily extended with new forecasting models or analytical features.

3. Data Processing Module

The first stage of the application is responsible for preparing raw transaction data for analysis.

The module performs:

- Reading historical Amazon transaction data
- Removing duplicate records
- Handling missing values
- Standardizing date formats
- Validating revenue calculations
- Converting transaction-level data into monthly aggregated data

The processed dataset serves as the foundation for forecasting and business analytics.

4. Feature Engineering

To improve forecasting accuracy, additional time-series features are generated.

Calendar Features

- Year
- Month
- Quarter

These features help models identify yearly business cycles.

Lag Features

Previous observations are used as predictive variables.

Implemented lag features include:

- Lag 1
- Lag 3
- Lag 6
- Lag 12

These enable machine learning models to learn temporal dependencies.

Rolling Statistics

The application calculates:

- Rolling Mean
- Rolling Standard Deviation

Rolling statistics reduce short-term fluctuations and capture long-term trends.

5. Hierarchical Forecasting Engine

Instead of forecasting every product using a single algorithm, the application follows a hierarchical forecasting strategy.

Level 1 – Category Forecasting

Each product category is forecast independently.

Models evaluated:

- Prophet
- Holt-Winters
- SARIMA

Category-level forecasting provides stable long-term demand estimates because categories have sufficient historical observations.

Level 2 – Top Product Forecasting

Products with high sales volume are forecast individually.

Models used:

- Prophet
- Holt-Winters

These products generally contain enough historical information for accurate statistical forecasting.

Level 3 – Remaining Product Forecasting

Products with limited historical observations are forecast using machine learning.

Algorithm:

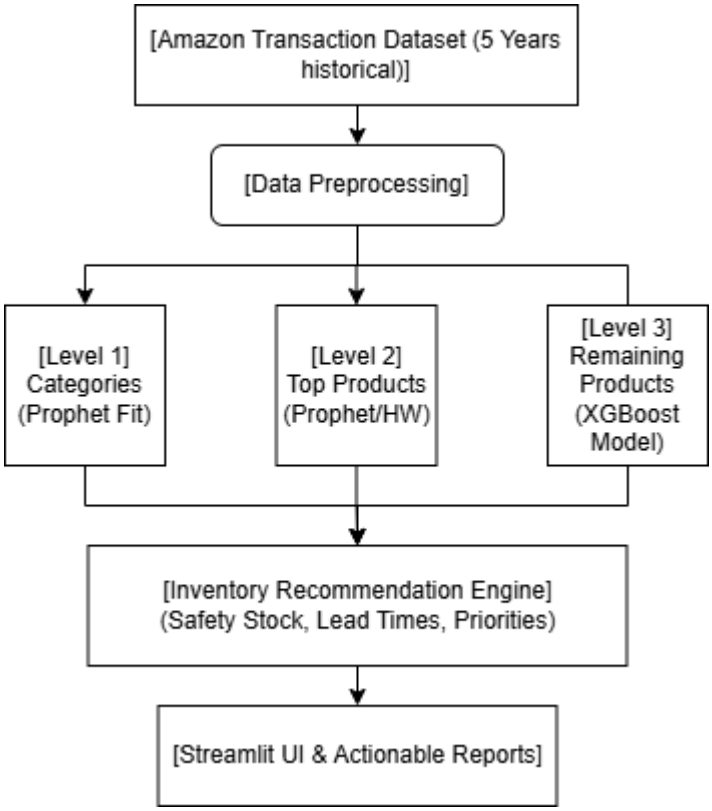
XGBoost Regressor

Input Features:

- Lag Features

- Month
- Quarter
- Rolling Mean
- Historical Sales
- Category

This approach provides greater scalability for products with sparse historical data.



6. Forecast Model Evaluation

Rather than assuming one model is best, the application evaluates multiple forecasting algorithms.

Evaluation metrics include:

MAE (Mean Absolute Error)

Measures the average forecasting error.

Lower values indicate better performance.

RMSE (Root Mean Squared Error)

Penalizes larger forecasting errors more heavily.

Used to compare forecasting accuracy.

MAPE (Mean Absolute Percentage Error)

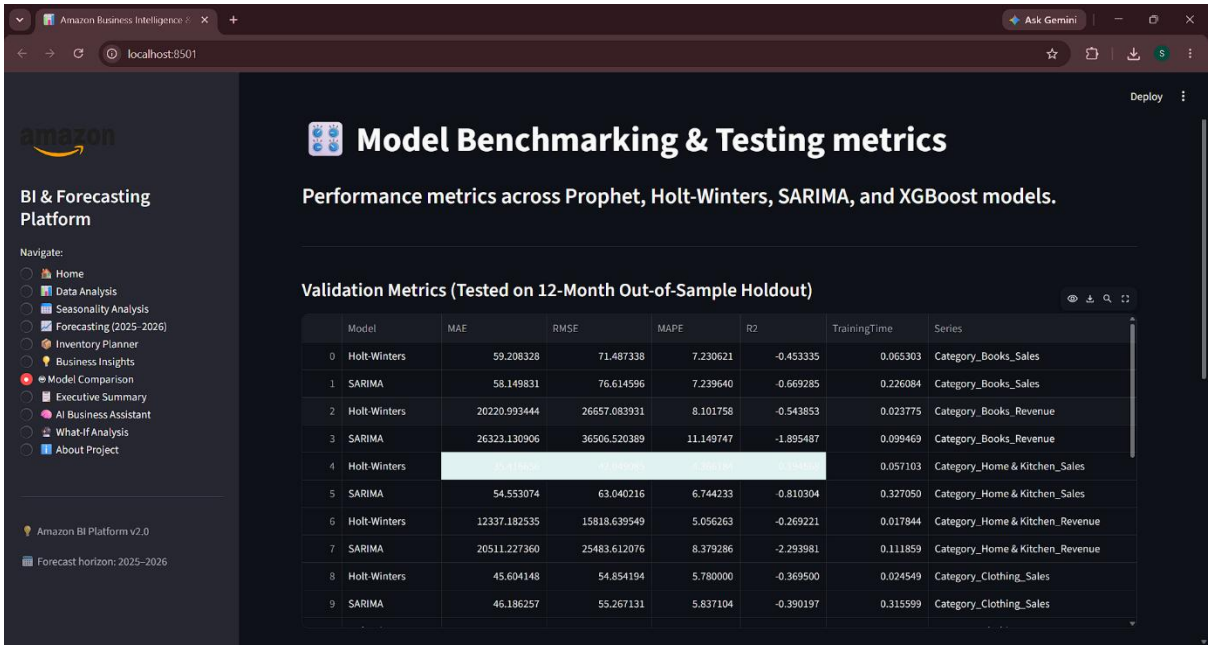
Measures forecasting error as a percentage.

Provides business-friendly interpretation of forecasting accuracy.

R^2 Score

Measures how well the model explains variation in demand.

Higher values indicate better model performance.



	Model	MAE	RMSE	MAPE	R2	TrainingTime	Series
0	Holt-Winters	59.208328	71.487338	7.230621	-0.453335	0.065303	Category_Books_Sales
1	SARIMA	58.149831	76.614596	7.239640	-0.669285	0.226084	Category_Books_Sales
2	Holt-Winters	20220.993444	26657.083931	8.101758	-0.543853	0.023775	Category_Books_Revenue
3	SARIMA	26323.130906	36506.520389	11.149747	-1.895487	0.099469	Category_Books_Revenue
4	Holt-Winters					0.057103	Category_Home & Kitchen_Sales
5	SARIMA	54.553074	63.040216	6.744233	-0.810304	0.327050	Category_Home & Kitchen_Sales
6	Holt-Winters	12337.182535	15818.639549	5.056263	-0.269221	0.017844	Category_Home & Kitchen_Revenue
7	SARIMA	20511.227360	25483.612076	8.379286	-2.293981	0.111859	Category_Home & Kitchen_Revenue
8	Holt-Winters	45.604148	54.854194	5.780000	-0.369500	0.024549	Category_Clothing_Sales
9	SARIMA	46.186257	55.267131	5.837104	-0.390197	0.315599	Category_Clothing_Sales

7. Forecast Generation

After model evaluation, the best-performing model generates forecasts for:

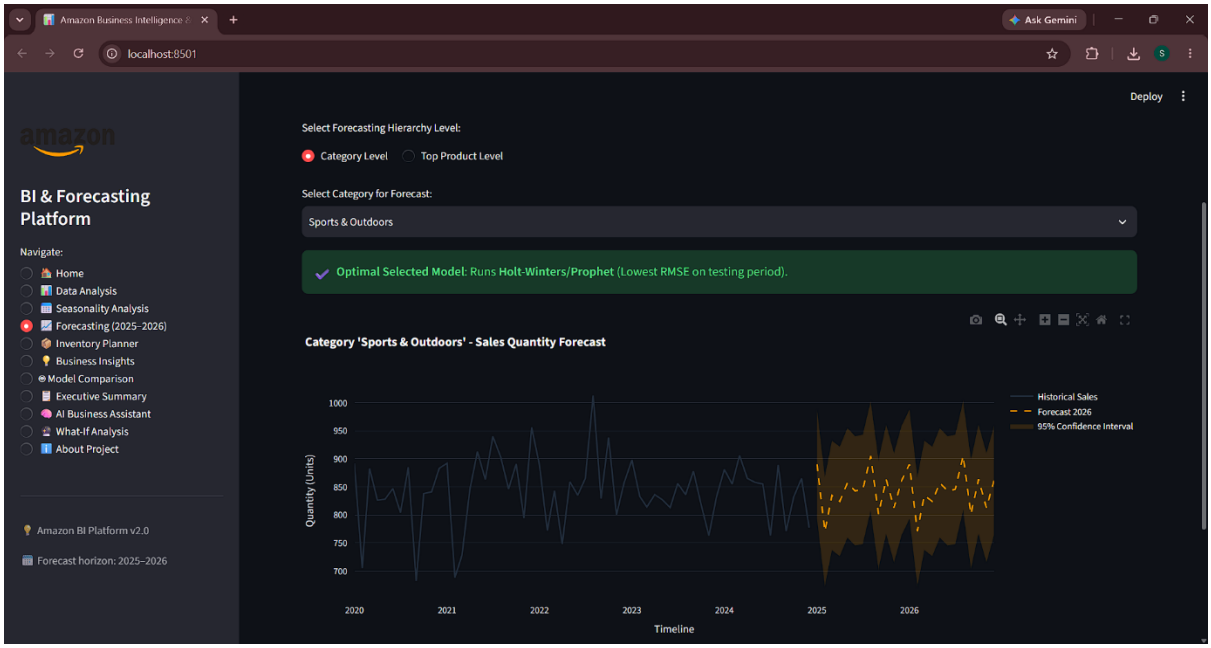
- Category Level
- Top Product Level
- Remaining Products

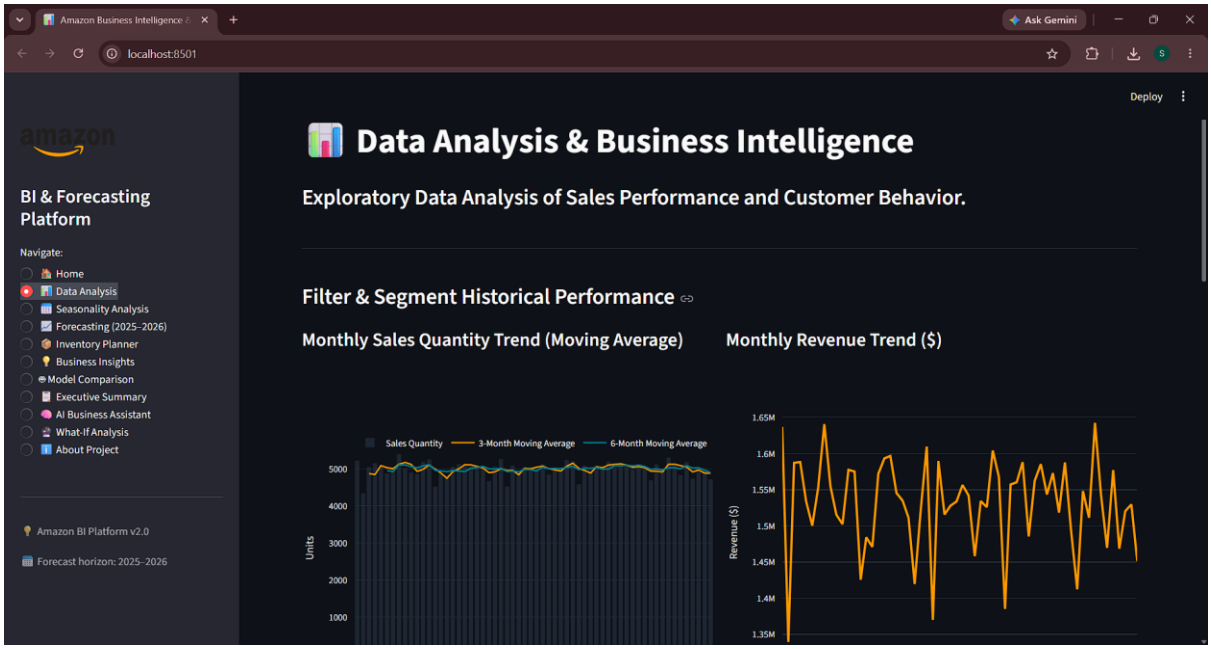
Forecasts are produced for:

- 2025
- 2026

Outputs include:

- Monthly Sales Forecast
- Revenue Forecast
- Confidence Intervals


Amazon Business Intelligence
Ask Gemini
localhost:8501
Deploy
amazon
BI & Forecasting Platform
Navigate:
Home
Data Analysis
Seasonality Analysis
Forecasting (2025–2026)
Inventory Planner
Business Insights
Model Comparison
Executive Summary
AI Business Assistant
What-If Analysis
About Project
Amazon BI Platform v2.0
Forecast horizon: 2025–2026
Select Forecasting Hierarchy Level:
Category Level
Top Product Level
Select Category for Forecast:
Sports & Outdoors
Optimal Selected Model: Runs Holt-Winters/Prophet (Lowest RMSE on testing period).
Category 'Sports & Outdoors' - Sales Quantity Forecast
Historical Sales
Forecast 2026
95% Confidence Interval
Quantity (Units)
Timeline


Amazon Business Intelligence
Ask Gemini
localhost:8501
Deploy
amazon
BI & Forecasting Platform
Navigate:
Home
Data Analysis
Seasonality Analysis
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Inventory Planner
Business Insights
Model Comparison
Executive Summary
AI Business Assistant
What-If Analysis
About Project
Amazon BI Platform v2.0
Forecast horizon: 2025–2026
Data Analysis & Business Intelligence
Exploratory Data Analysis of Sales Performance and Customer Behavior.
Filter & Segment Historical Performance
Monthly Sales Quantity Trend (Moving Average)
Monthly Revenue Trend ($)
Sales Quantity
3-Month Moving Average
6-Month Moving Average
Units
Revenue ($)

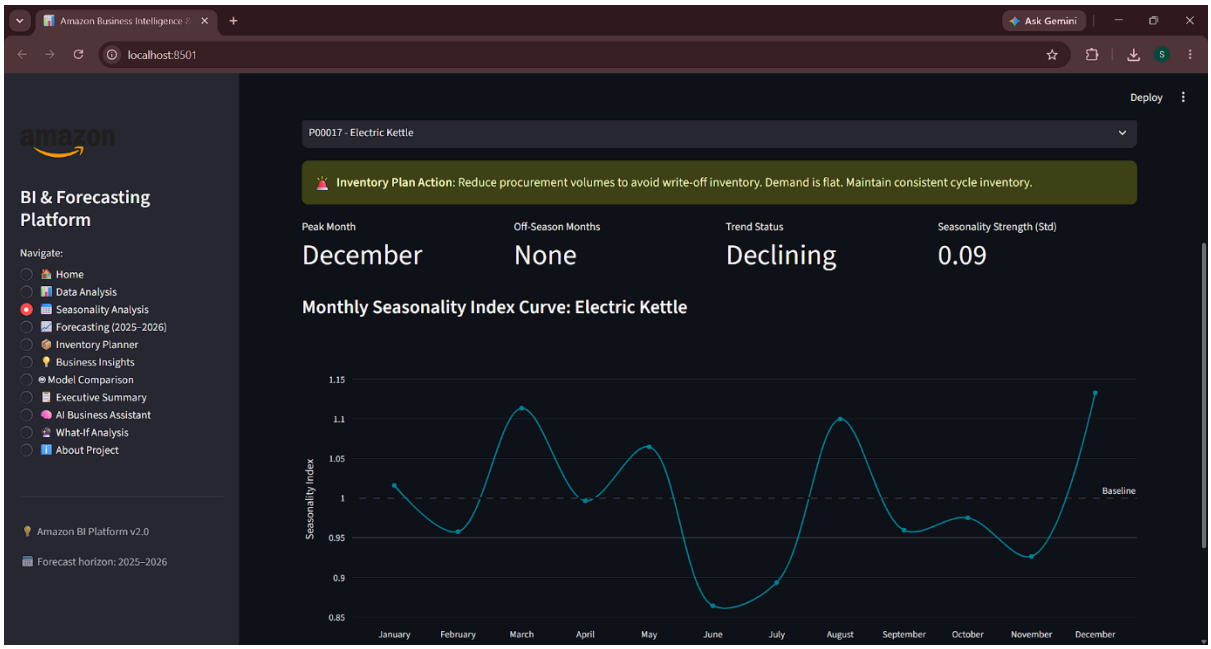
8. Seasonality Detection Engine

The Seasonality Engine analyzes historical monthly demand to identify recurring purchasing patterns.

The engine automatically determines:

- Peak demand months
- Off-season months
- Monthly demand distribution
- Seasonal demand strength
- Year-over-year growth

These insights enable proactive procurement planning before seasonal demand increases.



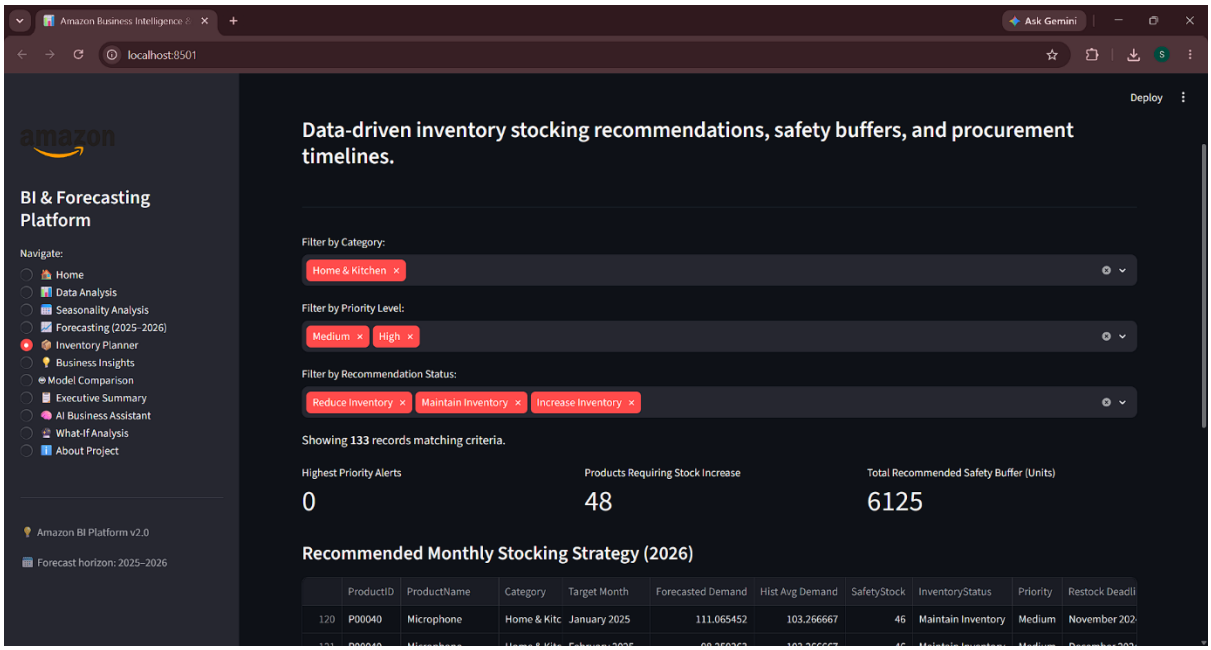
9. Inventory Optimization Engine

Forecast outputs are converted into inventory recommendations.

The Inventory Engine calculates:

- Forecast Demand
- Safety Stock
- Inventory Priority
- Procurement Month
- Business Risk
- Recommended Inventory Action

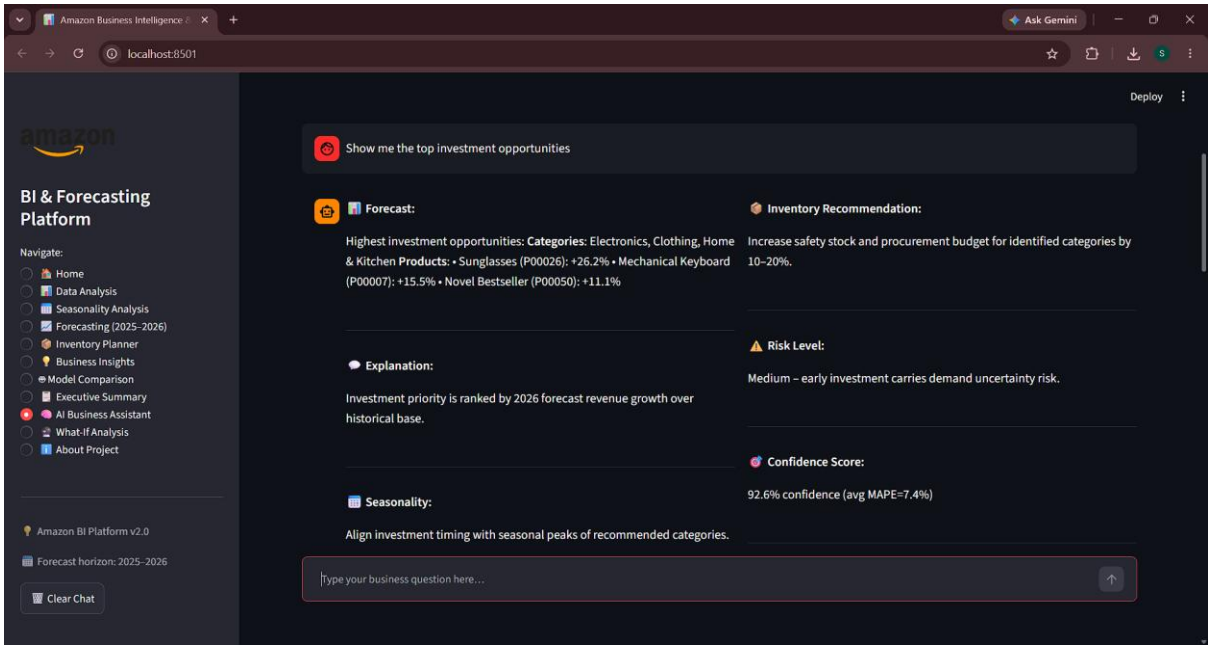
This enables inventory planning based on predicted future demand rather than historical averages.



10. Business Assistant Module

The AI Business Assistant is an intelligent query interface that enables users to interact with the analytical outputs generated by the platform. It allows business users to obtain insights related to sales performance, demand forecasting, seasonality, inventory planning, customer analytics, and executive summaries through natural language queries.

Instead of manually navigating multiple dashboards or reports, users can ask business-related questions, and the assistant retrieves relevant information from the processed datasets and generated analytical outputs to produce meaningful responses.



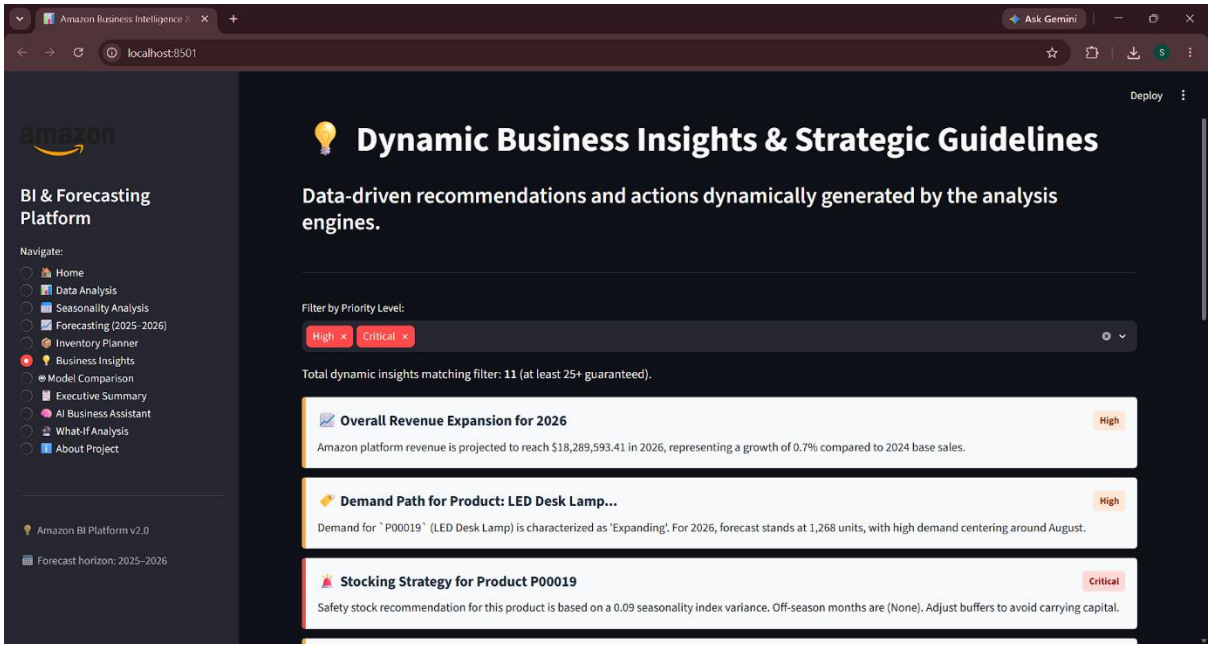
10. Business Recommendation Engine

The Business Recommendation Engine translates analytical outputs into actionable business decisions.

Examples of recommendations include:

- Increase inventory before forecasted peak demand.
- Prioritize investment in high-growth categories.
- Reduce procurement for declining products.
- Schedule purchasing based on seasonal demand.

The objective is to bridge the gap between forecasting results and business decision-making.

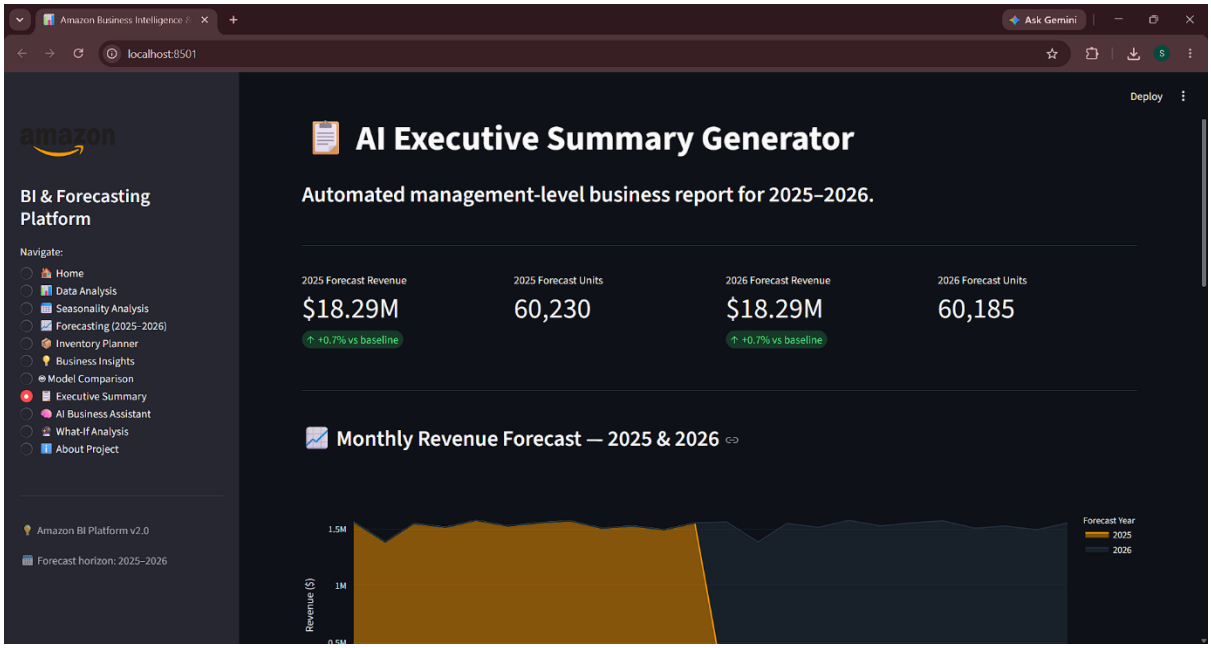


11. Executive Summary Engine

The application automatically generates executive-level insights, including:

- Highest Revenue Category
- Fastest Growing Category
- Top Performing Products
- Inventory Priorities
- Seasonal Opportunities
- Procurement Recommendations

This allows management to understand business performance without reviewing detailed analytical reports.



12. Business Intelligence Dashboard

Power BI dashboards provide interactive visualization of analytical results.

Dashboard modules include:

- Executive Dashboard
- Customer Analytics
- Product Performance
- Geographic Analysis
- Customer Sentiment

The dashboards support filtering by category, region, and time period, enabling interactive business exploration.

13. Application Outputs

The application generates the following outputs:

- Historical Business Analysis
- Category Performance Reports
- Product Performance Reports
- Forecast Reports (2025–2026)
- Inventory Planning Reports
- Seasonality Reports

- Executive Summary Reports
- Business Recommendations

14. Key Functionalities

- Automated data preprocessing
- Time-series feature engineering
- Hierarchical demand forecasting
- Multi-model forecasting evaluation
- Seasonal demand analysis
- Inventory planning
- Procurement recommendations
- Executive summary generation
- Interactive Power BI dashboards
- AI-assisted business insights

15. Conclusion

This project demonstrates an end-to-end Business Intelligence and Demand Forecasting solution that integrates data preprocessing, hierarchical forecasting, seasonality detection, and inventory optimization into a single analytical platform. The generated insights support informed business decisions and proactive inventory planning.